1. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

   Cross-Tabulation, Column Proportions
   happy * sex
   Data Frame: gss24

sex	male	female	Total
happy			
very happy	293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy	830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy	330 (22.7%)	370 (20.4%)	700 (21.5%)
Total	1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

   (a) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.
   (b) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
   (c) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.
   (d) The frequency of being `not too happy` is equal for men and women.

2. Which RStudio pane is used to write and edit R scripts and Quarto documents?

   (a) Console
   (b) Environment
   (c) Source
   (d) Files

3. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:

freq	median	mean	sd
3291	3	2.92	1.84

   (a) positively skewed
   (b) negatively skewed
   (c) nominal
   (d) symmetrical

4. The level of measurement for the `hapmar` variable is _________

   [1] "Happiness of marriage" Frequencies
   gss24$hapmar
   Type: Factor

	Freq	%	% Cum.
very happy	802	59.36	59.36
pretty happy	504	37.31	96.67
not too happy	45	3.33	100.00
Total	1351	100.00	100.00

(a) dichotomous
(b) interval
(c) ordinal
(d) nominal

5. Which command installs a new package in R?

   (a) `load()`
   (b) `require()`
   (c) `library()`
   (d) `install.packages()`

6. What is the result of the following code?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

   (a) The original gss24 dataset is overwritten.
   (b) A new column is added to `gss24`
   (c) rows with `age` > 20 are removed
   (d) Only rows with `age` > 20 and columns `age` and `childs` are returned.

7. What does the `::` operator do in R?

   (a) Comments out a line of code
   (b) Specifies a function from a particular package (without loading it)
   (c) Assigns a value to a variable
   (d) Imports all functions from a package

8. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?

   (a) percentages
   (b) cumulative frequencies
   (c) frequencies
   (d) cumulative percentages

9. In order to compare meaningfully across groups, a frequency should be converted into a/an ____________.

   (a) number
   (b) ordinal variable
   (c) standard deviation
   (d) percentage

10. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data?*

   (a) Statistics explain why relationship exist between variables
   (b) Statistics simplify complex information to help us make better decisions
   (c) Statistical formulas are the most important part of understanding data
   (d) Statistics provide honest answers to complex questions

11. Which of these datasets has a standard deviation of 0?

   (a) 0, 2, 4, 6, 8
   (b) 2, 2, 5, 7, 7
   (c) 1, 4, 7, 10, 15
   (d) 4, 4, 4, 4, 4

12. What does Oscar Gandy suggest about the use of racial statistics?
    (a) Their use in legal proceedings increasingly lead to unfair policy outcomes
    (b) They are neutral tools most useful for academic research
    (c) Their use in data journalism helps mitigate online polarization
    (d) They can both expose and perpetuate discrimination

13. What does the pipe operator (|>) do in dplyr?
    (a) It passes the result of one function into the next.
    (b) It loads the dplyr package into memory.
    (c) It filters rows based on a condition.
    (d) It creates a new column in a data frame.

14. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.53	7.93

    (a) The z-score is negative because 5 is below the sd.
    (b) The z-score is large and positive because 5 is much higher than the mean.
    (c) The z-score is close to zero because 5 is close to the mean.
    (d) The z-score is undefined because the median is 0.

15. Which function is used to load a package in R?
    (a) `load()`
    (b) `source()`
    (c) `library()`
    (d) `install.packages()`

16. What will happen if you try to use a function from a package that hasn't been loaded?
    (a) The function will run normally
    (b) R will automatically load the package
    (c) R will prompt you to install the package
    (d) You will get an error saying the function is not found

17. What does the expression `5 != 3` evaluate to in R?
    (a) TRUE
    (b) NA
    (c) FALSE
    (d) NULL

18. Which of the following is a measure of the spread of a distribution?
    (a) median
    (b) mean
    (c) standardized variable
    (d) standard deviation

19. In a normally distributed set of test scores, a student scoring below the average will have a _______________.
    (a) positive z-score
    (b) negative standard error
    (c) positive standard error
    (d) negative z-score

20. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
    (a) This couple had fewer arguments than about 2% of couples.
    (b) This couple had fewer arguments than about 98% of couples.
    (c) This couple had fewer arguments than about 5% of couples.
    (d) This couple had about 8 arguments per month.

21. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?
    (a) 50%
    (b) 62%
    (c) 95%
    (d) 68%

	(a)	(b)	(c)	(d)
1.			X	
2.			X	
3.		X		
4.			X	
5.				X
6.				X
7.		X		
8.				X
9.				X
10.		X		
11.				X
12.				X
13.	X			
14.			X	
15.			X	
16.				X
17.	X			
18.				X
19.				X
20.		X		
21.				X